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Hyperconvexity and domains of discontinuity for surface groups in $SL_3\mathbb{C}$

BY

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THESIS [PhD candidates may use DISSERTATION here instead]

Submitted as partial fulfillment of the requirements
for the degree of Doctorate of Philosophy in Mathematics
in the Graduate College of the
University of Illinois Chicago, 2026

Chicago, Illinois

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Dedication

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Acknowledgements

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Preface

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CHAPTER 1

Introduction

This is the introduction chapter. We cite some classic works [**?**, **?**].

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THEOREM 0.1. *This is a theorem*

We reference Theorem **??**.

University of Illinois Chicago.

1. Motivation

1.1. Historical context. A brief overview of how the problem developed.

$$\int_0^1 f(x)dx = 2 \tag{1}$$

How to solve (**??**)

1.2. Open questions. Some questions remain open for future work.

Note: I have not tested the accessibility of this table.

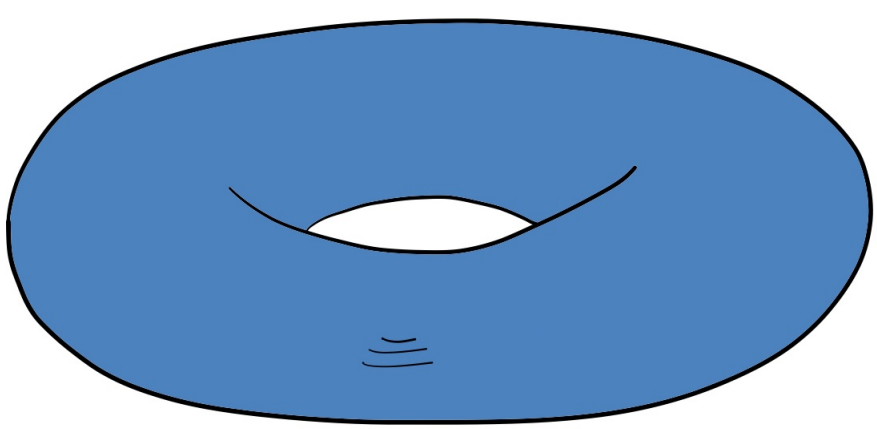

FIGURE 1. This is a torus

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & A' \arrow[r, "f'"] \arrow[d, "a"'] & B' \arrow[r, "g'"] \arrow[d, "b"'] & C' \arrow[r] \arrow[d, "c"'] & 0 \\
0 \arrow[r] & A \arrow[r, "f"] \arrow[d, "a''"'] & B \arrow[r, "g"] \arrow[d, "b''"'] & C \arrow[r] \arrow[d, "c''"'] & 0 \\
0 \arrow[r] & A'' \arrow[r, "f''"] \arrow[d] & B'' \arrow[r, "g''"] \arrow[d] & C'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

FIGURE 2. The Snake Lemma

Monkeys	Lions
100	200

TABLE 1. Example Table

CHAPTER 2

Background

This chapter gives necessary background.

1. Group theory

A group is a set G with a binary operation satisfying closure, associativity, identity, and inverses.

THEOREM 1.1. *Every finite subgroup of the multiplicative group of a field is cyclic.*

PROOF. This is a standard result from algebra. $\square$

CHAPTER 3

Main Results

Here we present the main contributions of the thesis.

1. A computer simulation

[basicstyle = ,resetmargins=true,tabsize=5,extendedchars=false] def factorial(n): """Compute the factorial of n recursively.""" if n == 0: return 1 else: return n * factorial(n - 1)

```
print(f"5!  = factorial(5)")
```

2. Second main result

Another significant theorem.

CHAPTER 4

Vitae

For UIC this can be a full CV or a short version (e.g. half a page) with previoius degrees, work experience and publications.

APPENDIX A

Technical Lemmas

Here we collect some supporting lemmas.

Bibliography

[1] R. Hartshorne, *Algebraic Geometry*, Springer-Verlag, New York, 1977.

[2] D. Mumford, *Abelian Varieties*, Oxford University Press, 1970.

[3] J. Draisma, E. Horobet, G. Ottaviani, B. Sturmfels, and R. R. Thomas, "The Euclidean distance degree of an algebraic variety," *arXiv:1309.0049* (2013). Available at: https://arxiv.org/abs/1309.0049